

20260217EB_MN (TDP43, ATP, IF, RNA, Hylight test)

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2026年2月17日 星期二

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 3.5 wells of 6-wells in two 11mL T2 10-cm cell-repellent plates
Dish A and B were processed separately to have two biological replicates
keep some iPSCs

2026年2月19日 星期四

T2 -> T3

2026年2月21日 星期六

T3 -> T4

2026年2月23日 星期一

T4 -> T5

2026年2月25日 星期三

T5 -> T6

2026年2月27日 星期五

T6 -> T6 (~14mL for over-weekend)
Coat PLO at RT

2026年2月28日 星期六

Coat Laminin at 37C

2026年3月2日 星期一

T6 -> T7

Dissociation (process dishA and dishB separately)

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 4mL 0.25% trypsin (prepare 8mL 0.25% trypsin with 100U DNaseI)
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette*10 at 5min, 10min, 15min
5. add 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times

6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

cell info			
	A	B	C
1	cell	passage	well#
2	WT11	P2	3.5

Table1		
	A	B
1		volume (mL)
2	trypsin	4
3	HI-FBS/CTWM	8

Table3										
	A	B	C	D	E	F	G	H	I	J
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	dish A	1.35	99	5	6.75	A	24	225	24	4000
3	dish B	2.07	99	5	10.35	B	24	225	6	1000
4						C	8-well chamber slide	100	8	593
5						D	6 (A1-3)	1217	3	the rest
6							6 (B1-3)	1967	3	

Well2						
	1	2	3	4	5	6
A			180 2uL	181 5uL		
B						
C						
D						

2026年3月3日 星期二

Change T7 -> T7 if added virus

2026年3月4日 星期三

T7-> T7 if not added virus

2026年3月5日 星期四

T7 -> T8

2026年3月7日 星期六

1/2 T8 -> 1/2 T8

2026年3月9日 星期一

1/2 T8 -> 1/2 T8

2026年3月11日 星期三

1/2 T8 -> 1/2 T8

Imaging TDP43 (hypoglycemia with LMB)

Since last time I got some weird results with one experiment of TDP43 and IF (I somehow suddenly had much higher N/C ratio with hypoglycemia), I want to prepare new media today.

I will treat the cells in the microscope incubator

1. Treat SPY505 (1:5000) at 8:30am
 - a. $500 \times 6 = 3000 / 5000 = 0.6$
 - b. 12uL BPT8 media + 0.6uL SPY505 -> 2uL/well
2. Prepare media
 - a. 2.5mM gluc: $14 \times 1.5 + 0.8 \times 2 = 21 + 1.6 = 22.6$
 - b. 0.15mM gluc: $16 \times 1.5 + 0.8 \times 4 = 24 + 3.2 = 27.2$
3. Prepare **JFX650**
stock: 2.5uM
working: 1nM
 - a. add 1:2500 into 2.5mM or 0.15mM glucose media
 - I. 0.28 in 700uL 2.5mM gluc
 - II. dilute 1:20 first (in 2.5mM and 0.15mM respectively), and add 5.6uL into 700uL aliquot
4. Prepare LMB:
 - a. 1.3uL in 700uL
 - b. Control: 1.3uL 70%MeOH in 700uL

Table2																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		stock	working	dilution	# of well	media for each well (uL)	total volume (uL)	drug volume (uL)									
2	Leptomycin B	5.6 mg/mL	10 ng/mL	560	1	700	700	1.3									
3	MeOH							1.3									

time table for treatment 2026/3/11														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	condition#			treat#1		treat#2		treat#3		treat#4		treat#5		collect
2	6-well			0h		1h		4h		5h		8h		9h
3	1	x6	2.5mM glucose constant	2.5mM gluc		2.5mM gluc		2.5mM gluc		2.5mM gluc	17:00	2.5mM gluc	20:00	21:00
4	2		0.15mM glucose recurrent	0.15mM gluc		2.5mM gluc		0.15mM gluc		2.5mM gluc		0.15mM gluc		
5	3		0.15mM glucose constant	0.15mM gluc	12:00-12:10	0.15mM gluc	13:00-13:10	0.15mM gluc	16:05-16:12	0.15mM gluc		0.15mM gluc		
6					SPY505	treat	1h image	JFX650	24h image on 3/12					
7	TDP43			treatment										
8	4	D1	2.5mM+70%MeO H	2.5mM gluc	8:40	9:50	11:12		10:10					
9	5	D2	0.15mM+70%MeO H	0.15mM gluc	8:40	10:05	11:20		10:23					
10	6	D3	0.15mM+LMB18.5 nM		8:40	10:15	11:27		10:46	There was bubbles so the time point was delayed...				
11		D4	2.5mM+70%MeO H	2.5mM gluc	8:40	10:30	11:35		11:04					
12		D5	0.15mM+70%MeO H	0.15mM gluc	8:40	10:40	11:47		11:15					
13		D6	0.15mM+LMB18.5 nM		8:40	10:50	11:54	9:05-9:10	11:30					
14														

20260311_timetable													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		D1		D2		D3		D4		D5		D6	
2	8:30	treat SPY505		treat SPY505		treat SPY505							
3	9:30	treat#1	2.5mM +JFX										
4	9:45			treat#1	0.15mM +JFX								
5	10:00					treat#1	0.15mM+LMB +JFX						
6	10:15							treat#1	2.5mM +JFX				
7	10:30	image@1 h								treat#1	0.15mM +JFX		
8	10:45			image@1 h								treat#1	0.15mM+LMB +JFX
9	11:00					image@1 h							
10	11:15							image@1h					
11	11:30									image@1h			
12	11:45											image@1h	

6-well with recurrent hypoglycemia

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	A	B
1		6-well
2	12:00	treat#1
3	12:30	
4	13:00	treat#2
5	13:30	
6	14:00	
7	14:30	
8	15:00	
9	15:30	
10	16:00	treat#3
11	16:30	
12	17:00	treat#4
13	17:30	
14	18:00	
15	18:30	
16	19:00	
17	19:30	
18	20:00	treat#5
19	3/12	
20	12:00	collect (24h)

TDP43_O.C. , 60X, 600x900, 16-bit

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	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	16	20	200 ms	
3	640	16	10	300 ms	

fileid_treat_TDP43

	A	B	C	D	E	F
1	image	treatment	treat_time	type	drug	replicate
2	MN_TDP43Halo_60x_0001	2.5mM gluc	1h	no-change	MeOH	A
3	MN_TDP43Halo_60x_0002	0.15mM gluc	1h	no-change	MeOH	A
4	MN_TDP43Halo_60x_0003	0.15mM gluc	1h	no-change	LMB-18.5nM	A
5	MN_TDP43Halo_60x_0004	2.5mM gluc	1h	no-change	MeOH	B
6	MN_TDP43Halo_60x_0005	0.15mM gluc	1h	no-change	MeOH	B
7	MN_TDP43Halo_60x_0006	0.15mM gluc	1h	no-change	LMB-18.5nM	B
8	MN_TDP43Halo_60x_0007	2.5mM gluc	24h	no-change	MeOH	A
9	MN_TDP43Halo_60x_0008	0.15mM gluc	24h	no-change	MeOH	A
10	MN_TDP43Halo_60x_0009	0.15mM gluc	24h	no-change	LMB-18.5nM	A
11	MN_TDP43Halo_60x_0010	2.5mM gluc	24h	no-change	MeOH	B
12	MN_TDP43Halo_60x_0011	0.15mM gluc	24h	no-change	MeOH	B
13	MN_TDP43Halo_60x_0012	0.15mM gluc	24h	no-change	LMB-18.5nM	B

2026年3月12日 星期四

Treat JFX650 at 9:00am
400/2500 = 0.16
0.16*2*20 = 6.4uL 2.5mM +0.32uL JFX650
0.16*4*20 = 12.8uL 0.15mM + 0.64uL JFX650

Collect 24h RNA sample at 12pm in 1mL Trizol

2026年3月13日 星期五

Image ATP/ADP ratio with different concentrations of glucose

Well1

	1	2	3	4	5	6
A	7	8			7	8
B	4	5	6	4	5	6
C	1	2	3	1	2	3
D						

prepare media with different conc. of glucose1

	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1	1100	20	55	3000	54.5	2945.5
3	2	20	7.5	2.6666666667	2000	750.0	1250.0
4	3	20	2.5	8	3000	375.0	2625.0
5	4	2.5	0.5	5	2000	400.0	1600.0
6	5	2.5	0.15	16.6666666667	2000	120.0	1880.0
7	6	2.5	0.05	50	2000	40.0	1960.0
8	7	2.5	0.015	166.6666666667	2000	12.0	1988.0
9	8		0		2000		

10:20 start preparing media

time table for treatment 2026/3/1

	A	B	C	D	E	F	G	H	I	J
1			batch A					batch B		
2	condition#		gluc conc.	treat	image			gluc conc.	treat	image
3	1	C1	20	12:16	12:17		C4	20	14:25	15:27
4	2	C2	7.5	12:32	13:31		C5	7.5	14:45	15:45
5	3	C3	2.5	12:47	13:47		C6	2.5	14:57	15:58
6	4	B1	0.5	13:01	13:59		B4	0.5	15:15	16:15
7	5	B2	0.15	13:20	13:19		B5	0.15	15:30	16:30
8	6	B3	0.05	13:36	14:42		B6	0.05	15:48	16:47
9	7	A1	0.015	13:50	14:55		A5	0.015	16:01	17:00
10	8	A2	0	14:02	15:07		A6	0	16:16	17:19

O.C. , 60X, zoom1.5, 600x900, 16-bit, Ti2 (new scope)					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	
2	DIC				
3	488	16	20	200	
4	Dapi-ex-Green-em	16	10	200	signal was stonrg this time

fileid_treat_ATP				
	A	B	C	D
1	image	treatment	treat_time	replicate
2	vLH92_MN_60x-1xzoom_0001	20	1h	A
3	vLH92_MN_60x-1xzoom_0002	7.5	1h	A
4	vLH92_MN_60x-1xzoom_0003	2.5	1h	A
5	vLH92_MN_60x-1xzoom_0004	0.5	1h	A
6	vLH92_MN_60x-1xzoom_0005	0.15	1h	A
7	vLH92_MN_60x-1xzoom_0006	0.05	1h	A
8	vLH92_MN_60x-1xzoom_0007	0.015	1h	A
9	vLH92_MN_60x-1xzoom_0008	0	1h	A
10	vLH92_MN_60x-1xzoom_0009	20	1h	B
11	vLH92_MN_60x-1xzoom_0010	7.5	1h	B
12	vLH92_MN_60x-1xzoom_0011	2.5	1h	B
13	vLH92_MN_60x-1xzoom_0012	0.5	1h	B
14	vLH92_MN_60x-1xzoom_0013	0.15	1h	B
15	vLH92_MN_60x-1xzoom_0014	0.05	1h	B
16	vLH92_MN_60x-1xzoom_0015	0.015	1h	B
17	vLH92_MN_60x-1xzoom_0016	0	1h	B

Check A3 and A4

A3: had some GFP+ cells (not many) with okay signal, but there a lot of debris and cells doesn't look very healthy.

A4: even fewer GFP+ cells. A lot of debris.

IF

Treat 8 well chamber slide for 1h from 17:07 - 18:07

1250uL media with 0.5uL 2.5uM (1:2500) JFX650
wash with 120uL media, add 150uL media
Fix with 4% PFA -> PBS wash -> permeabilize with 0.2% Triton -> blocking with 10% FBS -> cold room

Table4

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	A	B	C	D
1	2.5mM gluc	2.5mM gluc	2.5mM gluc	2.5mM gluc
2	0.15mM gluc	0.15mM gluc	0.15mM gluc	0.15mM gluc

2026年3月19日 星期四

stain primary antibody with 20250224MN

Table7

✓

	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	400	0.8	488	B	1000	800	0.8
3	A	147	Importin-b1	m	1000	400	0.4	594	O	1000	800	0.8
4	B	C-31	RANGAP1	Rb	500	800	1.6	488	B			
5	B	146	Ran	m	500	800	1.6	594	O			
6	C	82	HNRNPA2B1	Rb	500	400	0.8	594	B			
7	C	147	Importin-b1	m	1000	400	0.4	594	O	1000	800	0.8

2026年4月8日 星期三

IF imaging

Most well in 8-well chamber detached. Also, there are aa lot of bacteria contamination. --> do not image